Self-Supervised Hyperspectral Super-Resolution From RGB With Score-Distilled Spectral Prior

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***Abstract*—Recovering hyperspectral images (HSIs) from RGB observations is a highly ill-posed problem due to severe spectral information loss. However, current methods either rely on costly paired RGB–HSI datasets that are difficult to obtain or on unpaired RGB–HSI data for spectral guidance, which increases training costs and often leads to physically distorted spectral reconstructions. To address these challenges, we propose a self-supervised score-distilled spectral prior (SDSP) framework, which consists of a wavelet-based cross-attention reconstruction network (WCAR-Net) as the generator and a pretrained spectral diffusion model as the data-driven spectral expert. Score distillation sampling (SDS) is applied to the diffusion model to compute spectral gradients, which are incorporated into the generator's loss to iteratively refine its predictions and produce high-quality hyperspectral reconstructions. In WCAR-Net, a cross-attention refinement module is designed to fuse spatial details with high-level semantic features, while wavelet-based feature decomposition preserves fine frequency-domain structures. Guided by the spectral expert provided by the diffusion model, the network is trained in a self-supervised and iterative manner, thereby eliminating the need for paired RGB–HSI data. Extensive experiments on multiple datasets demonstrate that our method significantly outperforms state-of-the-art approaches in both spectral fidelity and spatial reconstruction.**

***Index Terms*—Diffusion model, hyperspectral image (HSI), score distillation sampling (SDS), self-supervised framework, spectral super-resolution (SSR).**

I. Introduction

THE hyperspectral image (HSI) provides dense spectral information across hundreds of contiguous bands, enabling fine-grained material identification and precise scene understanding. It has been widely applied in various fields such as remote sensing [1], [2], [3], [4], medical imaging [5], and agriculture [6]. However, due to the inherent tradeoff between spatial and spectral resolution in hyperspectral imaging, it remains challenging to capture HSI with both high spectral fidelity and fine spatial detail. Moreover, hyperspectral imaging systems are often expensive and complex, which greatly limits their widespread application in real-world scenarios.

To overcome these limitations, software-based hyperspectral imaging approaches have recently emerged to obtain high-resolution HSI (HR-HSI), including fusion-based super-resolution, spatial super-resolution, and spectral super-resolution (SSR) techniques. Fusion-based super-resolution aims to generate an HR-HSI by fusing a low-spatial-resolution HSI (LR-HSI) with a co-registered high-resolution multispectral (HR-MSI) or RGB image [7], [8], [9], [10], [11]. Specifically, representative unsupervised architectures, such as EDIP-Net [12] and M2U-Net [13], have been proposed to alleviate the reliance on paired training data. Although fusion-based methods achieve strong performance, they depend on paired LR-HSI and HR-MSI data from identical scenes, which are difficult to acquire and limit their practical applicability. The spatial super-resolution provides an alternative strategy by reconstructing HR-HSI from its LR-HSI counterpart [14], [15], [16]. Recent remote sensing image super-resolution studies have explored efficient Transformer architectures to reduce computational complexity and uncertainty-aware optimization to improve reconstruction reliability [17], [18]. However, acquiring HSI is often costly and time-consuming. In contrast, RGB images are inexpensive and easy to capture. Consequently, recent research has shifted toward SSR, which aims to reconstruct HR-HSI directly from a single RGB image, providing a more practical and efficient alternative.

Deep-learning-based approaches have recently become the dominant paradigm in SSR. Existing methods can be broadly categorized into supervised and unsupervised/unpaired approaches. Supervised methods [see Fig. 1(a)] learn direct nonlinear mappings from large-scale pixel-aligned RGB–HSI pairs and achieve strong reconstruction performance. However, their effectiveness critically depends on accurately co-registered training data, which is often difficult and expensive to obtain in real-world scenarios. To alleviate this limitation, unsupervised and unpaired methods [see Fig. 1(b)] exploit unpaired hyperspectral data to impose spectral or statistical constraints on the reconstruction process. Repre-

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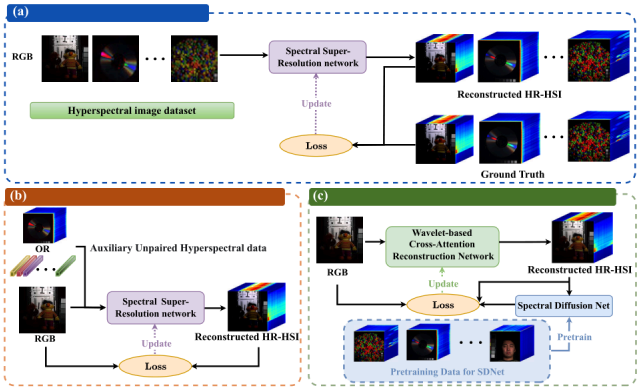


Fig. 1. Difference between existing deep-learning-based SSR methods and our self-supervised approach. (a) Supervised methods rely on large-scale paired RGB–HSI datasets for training. (b) Unsupervised or unpaired methods depend on auxiliary HSI data. (c) Our self-supervised SDSP framework integrates a pretrained spectral diffusion model with a generator WCAR-Net.

sentative approaches include unmixing-based constraints [19] and spectral point or structural priors [20]. Building upon these paradigms, recent works further incorporate external guidance, such as reference HSIs or spectral libraries, to enhance reconstruction fidelity [21], [22]. However, despite their improvements, these methods still rely on carefully selected reference HSIs, spectral libraries, or other fixed external priors, making their reconstruction quality dependent on the selected guidance. In addition, representative self-supervised methods exploit physical degradation models or spatial–spectral consistency constraints to reconstruct HSIs without ground-truth supervision [23], [24], [25], [26], [27], [28]. Nevertheless, these methods mainly rely on reconstruction fidelity losses for optimization and still require additional low-resolution HSIs from the same scene as the RGB image.

As illustrated in Fig. 1(c), the proposed score-distilled spectral prior (SDSP) framework introduces a diffusion-based spectral prior into the reconstruction process, allowing the spectral prior learned by a pretrained diffusion model to effectively guide self-supervised hyperspectral reconstruction. While the method requires pretraining on HSI data, it differs fundamentally from conventional paradigms by learning a continuous generative model of spectral distributions rather than relying on discrete reference exemplars or handcrafted spectral libraries. Specifically, the spectral diffusion network (SDNet) is first pretrained on unpaired HSIs to learn the continuous distribution of natural spectra. During the subsequent self-supervised reconstruction stage, the pretrained SDNet remains fixed and provides a learned spectral prior via score distillation sampling (SDS) to guide the optimization of wavelet-based cross-attention reconstruction network (WCAR-Net). As a result, the proposed framework performs SSR without paired RGB–HSI supervision or test-time reference images.

In fact, reconstructing HSIs from RGB images is a highly ill-posed inverse problem because much of the spectral information is lost during the imaging process. To tackle this challenge, our method is a synergistic dual-component framework comprising a pretrained SDNet and a self-supervised WCAR-Net. The SDNet acts as a learned spectral diffusion prior that learns the continuous distribution of natural spectra from unpaired HSIs rather than paired RGB–HSI data. Unlike conventional approaches that rely on pixel-aligned RGB–HSI supervision or test-time reference HSIs, SDNet provides a learned spectral expert prior that ensures spectrally faithful and stable reconstruction. Meanwhile, WCAR-Net exploits a wavelet-based architecture to capture multiscale spatial structures through cross-attention refinement, enabling effective reconstruction of spatial details. Overall, the proposed framework jointly leverages SDNet for spectral fidelity and WCAR-Net for spatial reconstruction, effectively addressing the ill-posed nature of SSR without paired supervision. Our contributions are summarized as follows.

1) We propose SDSP, a synergistic dual-component framework that couples a data-driven spectral prior learned from a pretrained diffusion model with a spatial–wavelet reconstruction network. To the best of the authors' knowledge, this is the first work to successfully leverage a pretrained score-based diffusion prior as guidance for self-supervised SSR, enabling effective optimization without paired RGB–HSI supervision.
2) We introduce a self-supervised WCAR-Net as an efficient spatial–wavelet generator, seamlessly coupled with a distillation mechanism that absorbs spectral knowledge from the SDNet. The WCAR-Net focuses on spatial fidelity, while the SDNet acts as a spectral expert to ensure spectrally faithful reconstruction.
3) We conduct extensive experiments demonstrating that our method outperforms even supervised methods, significantly narrowing the long-standing performance gap between unsupervised and supervised SSR. These results validate that a powerful learned prior can effectively replace paired-data supervision, establishing a new data-agnostic paradigm for high-fidelity HSI reconstruction.

II. Related Works

A. Spectral Super-Resolution

SSR enables the reconstruction of hyperspectral information from a single high-resolution RGB image. Early model-based methods, such as Arad [29] and A+ [30], primarily rely on handcrafted priors including sparse representation and low-rank decomposition. Although these methods possess good interpretability, their representation capability is limited, resulting in suboptimal reconstruction performance. With the advancement of deep learning, supervised models have become the dominant paradigm by learning complex nonlinear mappings from large-scale paired datasets. Representative architectures include convolutional networks such as HSCNN [31], HRNet [32], WHANet [33], SSR-Net [34], and frequency–spatial feature fusion networks [35], [36], as well as prior-embedded relational networks like HPRN [37] and multistage fusion frameworks, such as MSFN [38]. Despite their effectiveness, these methods rely heavily on paired supervision and exhibit limited generalization under domain shift. To reduce reliance on paired data, unsupervised and unpaired paradigms have been explored by leveraging cross-domain constraints and spectral priors. For instance, UnGUN [19]

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and MFormer [39] introduce unmixing-guided and masked modeling strategies, while SPointNet [20] leverages spectral sampling priors and Lib-SRDM [22] exploits spectral library-based semantic guidance. In addition, INR-HSISR [40] represents HSI as a continuous coordinate function for resolution-agnostic reconstruction.

Although these methods alleviate the reliance on paired supervision, they typically require carefully designed unsupervised constraints, auxiliary spectral priors, or external guidance to regularize the ill-posed reconstruction process, which may limit their robustness across different scenarios. More recently, Mamba-based state space models (SSRMamba [41], GMSR [42]) and Kolmogorov–Arnold networks (KANs) (NukesFormers [21]) have been explored to improve efficiency and representation capability for hyperspectral reconstruction. Nevertheless, these methods still mainly rely on progressively more sophisticated network architectures to improve reconstruction quality.

B. Diffusion and Wavelet-Based Super-Resolution

Diffusion and wavelet-based models have recently attracted considerable attention for their impressive performance in fusion-based super-resolution tasks. Diffusion-based methods typically condition on RGB or LR-HSI inputs and exploit the powerful generative capability of diffusion models to iteratively reconstruct HR-HSIs through denoising. For instance, HSR-Diff [43], ISPDiff [44], and S2CycleDiff [45] have been successfully applied to hyperspectral fusion, while others, such as SCDM [46], tackle the more challenging SSR problem. However, these paradigms remain limited, as they rely heavily on paired data for supervision and typically treat the diffusion model as a direct HR-HSI generator, restricting their generalization ability. In parallel, wavelet-based architectures decompose images into low- and high-frequency subbands, enabling networks to explicitly learn multiscale spatial representations that enhance spatial reconstruction quality. This strategy has shown strong effectiveness across diverse super-resolution tasks, including spatial super-resolution (MW-3D-convolutional neural network (CNN) [47] and WGSR [48]), image fusion (PF-MAWN [49]), and SSR (WHANet [33]). Nevertheless, despite their spatial advantages, these methods inherently lack mechanisms to preserve or enforce spectral consistency, leading to spectral distortion in reconstruction results.

III. Proposed Method

The goal of SSR is to generate an HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ from an observed RGB image $\mathcal{Y} \in \mathbb{R}^{H\times W\times 3}$ ($C \gg 3$). The HSI and RGB images are related through a spectral mapping process, formulated as

$$\mathcal{Y} = \mathcal{X} \times_3 \mathbf{R} + \mathcal{N} \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{3\times C}$ is the spectral response function (SRF) matrix, and $\mathcal{N}$ denotes the error term. The maximum a posteriori (MAP) estimate, $\hat{\mathcal{X}}$, aims to maximize $p(\mathcal{X}|\mathcal{Y})$, equivalently minimizing the negative log-posterior

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} -\log p\left(\mathcal{Y}|\mathcal{X}\right) - \log p\left(\mathcal{X}\right) \tag{2}$$

where $\log p(\mathcal{Y}|\mathcal{X})$ denotes the data likelihood of the observations $\mathcal{Y}$, and $\log p(\mathcal{X})$ represents the prior term of the HSI $\mathcal{X}$. The data likelihood term $\log p(\mathcal{Y}|\mathcal{X})$ can be further formulated as follows:

$$\mathcal{L}_{\text{data}}\left(\mathcal{X}\right) = \left\|\mathcal{X} \times_3 \mathbf{R} - \mathcal{Y}\right\|_F^2. \tag{3}$$

A. Overall Pipeline

Recent studies have shown that deep networks can capture complex structures in HSI. We propose a training-free WCAR-Net [see Fig. 2(b)] to reconstruct the HSI, effectively extracting global, local, and multiscale information from the RGB image. This network adaptively learns the deep prior of the HSI by leveraging both wavelet transforms and attention mechanisms, allowing for efficient feature extraction and multiscale representation learning. The mapping function can be expressed as follows:

$$\mathcal{X} = G_\zeta\left(\mathcal{Y}\right) \tag{4}$$

where ζ denotes the learnable parameters of $G_\zeta(\mathcal{Y})$.

Unfortunately, reconstructing HSI from an RGB image is a highly ill-posed problem due to severe spectral information loss. Using only the WCAR-Net generator in (4) and optimizing solely with the data fidelity term in (3) is insufficient to address this challenge, as confirmed by the ablation study in Table II. Therefore, various priors have been explored to constrain the solution space and enhance spectral reconstruction fidelity, including handcrafted priors for model-based approaches and auxiliary unpaired HSI priors for unsupervised or unpaired learning methods. However, handcrafted priors lack sufficient capacity to capture the intricate spatial–spectral structures of HSI, while auxiliary HSI priors depend heavily on external reference data, making the reconstruction sensitive to the content and spectral properties of the guidance.

Diffusion models [50] possess strong generative capabilities for learning complex high-dimensional data distributions, enabling the automated learning of image priors beyond manual design. In this work, we propose an SDSP, which serves as the powerful diffusion prior term $\log p(\mathcal{X})$ in (2). It distills diffusion-based spectral expert knowledge from a pretrained diffusion model into the self-supervised trainable generator $G_\zeta(\mathcal{Y})$, thereby enforcing spectral authenticity through gradient-based guidance. Overall, the proposed SDSP framework integrates a pretrained diffusion model, SDNet [see Fig. 2(c)], as a data-driven spectral expert with WCAR-Net [see Fig. 2(b)], which offers strong spatial–wavelet representation capability for SSR. Within this framework, WCAR-Net is responsible for recovering spatial details and capturing multiscale frequency information from the input RGB image, while SDNet provides a complementary spectral prior learned via diffusion-based generative modeling. The SDS mechanism serves as a bridge, progressively transferring the spectral knowledge of SDNet into the spatial reconstruction process of WCAR-Net, thereby enhancing both spatial reconstruction quality and spectral fidelity. The overall pipeline of the proposed SDSP framework is illustrated in Fig. 2.

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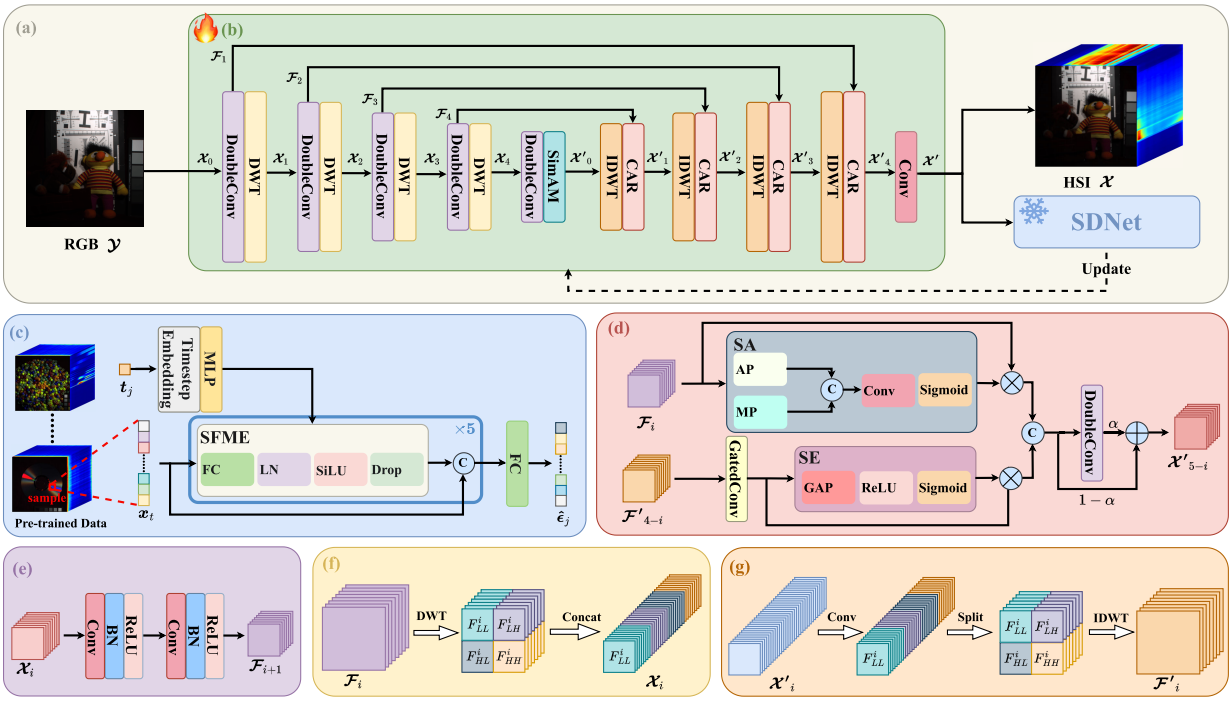


Fig. 2. Overview of (a) proposed SDSP framework, (b) composed of the WCAR-Net generator, (c) pretrained SDNet, (d) CAR, (e) DoubleConv, (f) DWT, and (g) IDWT.

TABLE I

QUANTITATIVE RESULTS OF COMPARISON METHODS. PSNR, SSIM, AND CC ARE THE HIGHER THE BETTER (↑), WHILE SAM, ERGAS, AND NIQE ARE THE LOWER THE BETTER (↓)

Dataset	Metric	Arad [29]	INR-HSISR [40]	HPRN [37]	UnGUN [19]	MSFN [38]	FSDFF [35]	NukesFormers [21]	Lib-SRDM [22]	Ours
CAVE	PSNR ↑	26.57	35.10	34.54	25.62	36.19	33.41	30.00	34.08	**36.45**
	SSIM ↑	0.847	0.967	0.961	0.844	0.976	0.938	0.887	0.944	**0.977**
	SAM ↓	21.87	11.34	12.57	29.64	**7.95**	12.87	14.82	14.72	11.15
	ERGAS ↓	15.17	6.00	5.72	18.09	4.93	7.13	10.48	6.31	**4.46**
	CC ↑	0.929	0.992	0.990	0.910	0.992	0.984	0.976	0.990	**0.993**
	NIQE ↓	0.351	0.217	0.168	**0.138**	0.165	0.172	0.298	0.202	0.143
ICVL	PSNR ↑	37.72	40.65	39.91	29.23	34.31	35.84	39.89	36.46	**41.05**
	SSIM ↑	0.966	0.983	0.984	0.941	0.981	0.948	0.982	0.962	**0.988**
	SAM ↓	4.65	4.02	4.42	10.08	3.98	6.39	5.01	7.81	**3.20**
	ERGAS ↓	4.04	**2.93**	3.51	26.64	3.70	4.31	3.48	4.58	3.88
	CC ↑	0.968	**0.986**	0.983	0.980	0.979	0.968	0.984	0.975	0.983
	NIQE ↓	0.244	0.183	0.180	0.189	0.165	0.229	0.190	0.255	**0.159**
PaviaU	PSNR ↑	30.19	28.97	25.80	28.54	32.12	31.65	28.52	**34.35**	32.61
	SSIM ↑	0.914	0.928	0.819	0.950	0.954	0.944	0.958	**0.967**	0.956
	SAM ↓	4.41	5.42	5.91	4.98	3.31	3.48	4.36	**2.81**	3.20
	ERGAS ↓	1.02	1.92	2.99	1.29	1.89	0.99	2.77	**0.74**	0.87
	CC ↑	0.987	0.986	0.986	0.987	0.989	0.988	0.982	**0.993**	0.990
	NIQE ↓	0.358	**0.314**	**0.314**	0.322	0.316	0.318	0.320	0.344	0.330
Chikusei	PSNR ↑	44.90	39.81	44.50	35.82	46.96	43.50	42.03	39.92	**47.34**
	SSIM ↑	0.987	0.969	0.974	0.935	0.993	0.987	0.979	0.944	**0.994**
	SAM ↓	3.25	4.40	3.74	8.12	2.61	2.78	4.37	7.41	**2.54**
	ERGAS ↓	4.25	5.80	3.56	8.56	**2.84**	3.07	4.45	6.35	3.30
	CC ↑	0.979	0.973	0.983	0.968	**0.988**	0.987	0.977	0.970	0.987
	NIQE ↓	0.356	0.328	0.310	0.360	0.302	0.332	0.321	**0.264**	0.304
QUH-Qingyun	PSNR ↑	42.41	29.04	43.53	27.97	44.11	34.74	33.14	41.51	**44.58**
	SSIM ↑	0.951	0.894	0.987	0.928	**0.991**	0.963	0.914	0.963	**0.991**
	SAM ↓	7.05	9.73	5.37	9.50	4.41	4.72	14.60	8.68	**2.01**
	ERGAS ↓	4.25	9.86	3.54	7.33	3.38	3.69	11.35	4.68	**0.77**
	CC ↑	0.932	0.893	0.962	0.901	0.964	0.948	0.672	0.943	**0.995**
	NIQE ↓	0.365	0.212	**0.082**	0.154	0.100	0.217	0.146	0.296	0.159

B. Score-Distilled Spectral Prior

The SDNet [51] serves as a generative network designed to capture the underlying data distribution of 1-D spectral data in HSI, which carry rich and physically meaningful information. Inspired by SDS [52], we introduce the SDSP, which transfers the spectral knowledge of a pretrained diffusion network into

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TABLE II
ABLATION STUDY RESULTS OF SDNET AND WCAR-NET

Dataset	Method	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	NIQE ↓
CAVE	w/o SDNet	20.16	0.799	41.26	35.94	0.891	0.209
	w/o WCAR	26.19	0.673	34.39	12.65	0.939	0.262
	Ours	**36.45**	**0.977**	**11.15**	**4.46**	**0.993**	**0.143**
QUH-Qingyun	w/o SDNet	14.36	0.526	56.37	62.16	0.726	0.181
	w/o WCAR	40.83	0.984	3.45	2.19	0.992	0.162
	Ours	**44.58**	**0.991**	**2.01**	**0.77**	**0.995**	**0.159**

our reconstruction model via score-based gradient guidance. This mechanism effectively transfers the rich spectral prior learned by the diffusion model to guide self-supervised SSR, providing a flexible and effective optimization framework.

The rationale for employing a 1-D spectral diffusion prior, rather than direct 3-D spatial–spectral learning, is motivated by the intrinsic ill-posed nature of the SSR problem. Due to the highly underdetermined mapping, directly modeling 3-D joint distributions often leads to spatial artifacts and spectral distortions. In contrast, the proposed SDNet focuses on modeling the statistical distribution of 1-D spectral signatures. This decoupling of spectral modeling from spatial complexity not only reduces computational cost but also enables the use of diffusion-based generative modeling to capture rich spectral priors. Through SDS, these learned priors are effectively transferred to the reconstruction process, ensuring high spectral fidelity even without paired training data.

1) Expert Prior: A Pretrained Diffusion Model: The proposed SDNet is developed based on the denoising diffusion probabilistic model (DDPM). It operates on 1-D spectral vector data and consists of two iterative stages: a forward spectral diffusion process and a reverse spectral generation process. Given a spectral vector $\mathbf{x} \in \mathbb{R}^C$ from the HSI, the spectral components of the HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ are assumed to be independent and identically distributed. The forward diffusion process can be formulated under the reparameterization trick as follows:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \tag{5}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a random noise, and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ is the cumulative product of the predefined single-step coefficient α_t. In the reverse spectral generation process, the model learns to reconstruct clean spectra from noisy observations by progressively denoising $\mathbf{x}_t$ to obtain $\mathbf{x}_{t-1}$. Specifically, the denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, parameterized by θ, is employed in SDNet to approximate the intractable posterior distribution $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ based on the input $\mathbf{x}_t$ and its corresponding diffusion timestep t

$$p\left(\mathbf{x}_{t-1}\middle|\mathbf{x}_t\right) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_\theta\left(\mathbf{x}_t, t\right), \sigma_t^2\mathbf{I}\right) \tag{6}$$

where μ_θ is the learnable mean, and σ_t is a predefined variance. The mean term $\mu_\theta(\mathbf{x}_t, t)$ can be rewritten as

$$\mu_\theta\left(\mathbf{x}_t, t\right) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta\left(\mathbf{x}_t, t\right)\right) \tag{7}$$

and $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ predicts the noise component in $\mathbf{x}_t$. The denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ of SDNet can be trained by optimizing the following simple denoising-based objective:

$$L_\theta\left(\theta\right) = \mathbb{E}_{\mathbf{x}_0,\boldsymbol{\epsilon},t}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|^2. \tag{8}$$

As shown in Fig. 2(c), the denoising network of the proposed SDNet, denoted as $\boldsymbol{\epsilon}_\theta(x_t, t)$, is inspired by [51] and built upon an efficient multilayer perceptron (MLP)-based architecture. Specifically, the SDNet is pretrained offline using spectral vectors sampled from available HSI datasets to capture the intrinsic physical distribution of natural spectra. In the SDSP framework, this pretrained SDNet is then kept frozen and acts solely as the external "expert prior."

2) Spectral Guidance via SDNet: We aim to reconstruct a high-quality HSI using the self-supervised WCAR-Net generator $\mathcal{X} = G_\zeta(\mathcal{Y})$, guided by spectral prior knowledge distilled from the pretrained diffusion model $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$. The diffusion model serves as a frozen spectral expert, providing effective prior guidance for hyperspectral super-resolution without requiring backpropagation. Specifically, the prior term $-\log p(\mathcal{X})$ in (2) constitutes the core of our SDSP, transferring the rich spectral expertise of the frozen SDNet into the trainable WCAR-Net generator $\mathcal{X} = G_\zeta(\mathcal{Y})$. In practice, this process is implemented through the prior loss $\mathcal{L}_{\text{prior}}$, which is evaluated at a single, randomly sampled timestep for each pixel and is defined as follows:

$$\mathcal{L}_{\text{prior}}\left(\mathbf{x}\right) = \mathbb{E}_{\boldsymbol{\epsilon}}\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x} + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t\right)\right\|^2. \tag{9}$$

During this iterative optimization process, the pretrained SDNet parameters θ remain frozen, acting as a static spectral expert to provide consistent gradient guidance without being updated by the reconstruction loss. Without loss of generality, we further incorporate the intrinsic spatial smoothness prior of HSI using total variation (TV) regularization. By jointly combining the data fidelity term in (3), the spectral prior term in (9) derived from the pretrained SDNet, and the untrained generative network $G_\zeta(\mathcal{Y})$ with TV loss, the overall optimization objective function of the proposed SDSP framework can be formulated as follows:

$$\min_{\zeta} \mathcal{L}_{\text{data}}\left(G_\zeta\left(\mathcal{Y}\right)\right) + \gamma\mathcal{L}_{\text{prior}}\left(G_\zeta\left(\mathcal{Y}\right)\right) + \lambda\mathcal{L}_{\text{TV}}\left(G_\zeta\left(\mathcal{Y}\right)\right)$$

where γ and λ are the balancing hyperparameters. Overall, the proposed SDSP framework comprises a pretrained SDNet diffusion model, which serves as an efficient frozen spectral expert providing prior guidance without backpropagation, and a WCAR-Net that functions as a powerful spatial–wavelet generator, effectively distilling spectral knowledge from the SDNet. The complete process of the proposed framework is summarized in Algorithm 1.

As illustrated in Fig. 2(b), the proposed WCAR-Net adapts a U-Net architecture [53], where DoubleConv and calibrated attention refinement (CAR) modules form the core building blocks of the encoder and decoder, respectively. Unlike conventional U-Nets that rely on pooling and upsampling, our network employs discrete wavelet transform (DWT) and its inverse (IDWT) [54] for downsampling and upsampling [see Fig. 2(f) and (g)]. This design not only preserves fine-grained spatial details through a lossless transformation, but also enables effective separation and integration of multi-scale global–local information by decomposing features into a low-frequency component (LL, global information) and three high-frequency components (LH, HL, HH, and local

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Algorithm 1 Optimization of the Proposed Framework

Input: Observed RGB $\mathcal{Y}$, SRF $\mathbf{R}$, clean HSI dataset $\tilde{\mathcal{X}}$, learning rate η, optimization steps K, diffusion time-steps T, and parameters γ and λ.
Train the denoising network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ in SDNet by randomly sampling spectral vectors from $\tilde{\mathcal{X}}$ and minimizing the objective defined in Eq. (8).
Fix SDNet parameters θ, **initialize** parameters ζ, set $k \leftarrow 0$, and let $\mathcal{X}^{(0)} = G_\zeta(\mathcal{Y})$.
while not convergence and $k < K$ **do**
  Sample time-step $t \sim U(1, \ldots, T)$
  Sample noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\mathcal{L}_{\text{prior}} = \sum_{\mathbf{x} \in \mathcal{X}^{(k)}} \mathbb{E}_{\boldsymbol{\epsilon}} \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x} + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\|^2$
  Loss $= \|\mathcal{X}^{(k)} \times_3 \mathbf{R} - \mathcal{Y}\|_F^2 + \gamma \mathcal{L}_{\text{prior}} + \lambda \mathcal{L}_{\text{TV}}(\mathcal{X}^{(k)})$
  Update parameters ζ via backpropagation using the Adam optimizer.
  $k = k + 1$ and $\mathcal{X}^{(k)} = G_\zeta(\mathcal{Y})$
  Check the convergence condition: $\frac{\|\mathcal{X}^{(k)} - \mathcal{X}^{(k-1)}\|_F}{\|\mathcal{X}^{(k-1)}\|_F} \leq 10^{-3}$.
end while
Output: HSI $\mathcal{X}^{(k)}$.

details). Specifically, the encoder progressively extracts multiscale hierarchical features from low-level spatial details to high-level semantics, where each DoubleConv block produces intermediate features $\mathcal{F}_i$, which are then decomposed by DWT into multifrequency representations $\mathcal{X}_i$ for effective separation and focused modeling of high- and low-frequency information. At the bottleneck, a parameter-free SimAM [55] module enhances semantic discriminability. In the decoder, features are upsampled via IDWT to obtain $\mathcal{F}'_i$ and then refined through CAR modules that fuse the corresponding encoder features $\mathcal{F}_i$, producing the final decoding representations $\mathcal{X}'_i$. The reconstructed HSI is obtained from the output of the last decoder block through a convolutional layer.

3) Calibrated Attention Refinement: To achieve effective feature aggregation between the encoder and decoder, we propose the CAR module, which goes beyond the simple concatenation or summation used in conventional U-Nets. In our CAR module, the skip-connection feature $\mathcal{F}_i$ from the DoubleConv block is processed by a spatial attention (SA) module to emphasize critical spatial details, whereas the decoder feature $\mathcal{F}'_i$ is obtained by IDWT of the feature $\mathcal{X}'_i$, followed by gated convolution and a squeeze-and-excitation (SE) module to effectively capture channel dependencies. This design is motivated by the complementary characteristics of the features: skip-connection features primarily preserve fine-grained spatial details, making SA ideal, while decoder features focus more on high-level semantic information, for which channelwise recalibration via SE efficiently enhances relevant semantic responses. Finally, the CAR output is generated via an adaptive residual fusion step, in which a learnable scalar α computes a weighted sum of the deeply processed features and the initially fused features. Overall, the skip-connection feature $\mathcal{F}_i$ and the decoder feature $\mathcal{F}'_{4-i}$ are adaptively fused, formulated as

$$\mathcal{X}'_{5-i} = f^i_{car}\left(\mathcal{F}_i, \mathcal{F}'_{4-i}\right) \tag{10}$$

where $f^i_{car}(\cdot)$ represents the CAR module in the ith layer. Notably, although the WCAR-Net is based on a CNN architecture, it is sufficiently capable of capturing the spatial features of HSI. Moreover, since our method is a self-supervised paradigm, the generator network should not be overly complex in order to maintain efficiency.

C. Discussion on Design Differences and Advantages

1) Comparison With Diffusion-Based Spectral Prior Methods: It is worth noting that, while Liu et al. [51] also employ a diffusion-based spectral prior, the proposed SDSP framework differs fundamentally in its optimization strategy and model formulation. Specifically, Liu et al. [51] directly optimize spectral variables in the pixel space through sequential refinement along the reverse diffusion chain. In contrast, SDSP optimizes the parameters of a structural generator, where the diffusion model serves as a frozen spectral prior and provides guidance via SDS with randomly sampled timesteps. This shift from pixel-space optimization to parameter-space optimization not only improves computational efficiency but also enables the integration of an implicit neural prior, leading to enhanced spatial–spectral consistency in the reconstruction.

2) Comparison With Traditional Frequency-Domain Methods: Recent studies have demonstrated that frequency-domain methods are highly effective for hyperspectral imaging tasks. Representative methods include MW-3D-CNN [47], FSDFF [35], PF-MAWN [49], WHANet [33], and WFA-SRNet [56], which exploit frequency-domain representations to preserve spatial structures and recover high-frequency details. However, SSR remains a highly ill-posed problem due to the severe loss of spectral information. Although existing frequency-domain methods effectively enhance spatial representation, they generally lack explicit spectral constraints, making the recovered spectra susceptible to spectral inconsistency. To address this issue, the proposed SDSP complements frequency-domain spatial modeling with a diffusion-based spectral prior. Specifically, WCAR-Net exploits frequency-domain representations to preserve multiscale spatial information, while SDNet learns the distribution of hyperspectral signatures and transfers this prior through SDS to guide reconstruction. Consequently, the reconstructed spectra are constrained to a physically meaningful spectral manifold while preserving rich spatial details. Therefore, the proposed framework jointly leverages frequency-domain spatial modeling and diffusion-based spectral prior learning, where WCAR-Net preserves spatial structures, and SDNet provides explicit spectral regularization for spectrally faithful reconstruction.

IV. Experiments

In this section, we evaluate the performance of our proposed SDSP against several state-of-the-art methods on five benchmark datasets. Moreover, we conduct a series of ablation studies to demonstrate the effectiveness of each key component within our proposed method.

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A. Experimental Setup

1) Datasets and Metrics: We evaluate the proposed method on CAVE, ICVL, Pavia University (PaviaU), Chikusei, and Qingdao University Hyperspectral (QUH-Qingyun), covering both natural-scene and remote sensing HSI datasets. CAVE contains 32 natural scenes with a spatial resolution of 512 × 512 pixels and a spectral range from 400 to 700 nm at 10-nm intervals (31 bands), and is widely used for evaluating reconstruction performance over diverse material reflectances. ICVL provides high-resolution images of 1392 × 1300 pixels with 31 contiguous spectral bands, serving as a standard benchmark for natural-scene SSR. PaviaU comprises 103 spectral bands with a spatial resolution of 1.3 m/pixel, representing complex urban structures. Chikusei is a large-scale airborne hyperspectral benchmark with 128 spectral bands ranging from 363 to 1018 nm, a spatial resolution of 2.5 m/pixel, and a spatial size of 2517 × 2335 pixels. QUH-Qingyun provides up to 270 spectral bands covering the 400–1000 nm range with ultrahigh spatial resolution, enabling a more nuanced assessment of fine-grained details across diverse land covers. These diverse datasets enable a comprehensive evaluation of spatial texture and spectral fidelity.

Following the standard simulation protocol for SSR, the input observations are generated from the ground-truth HSIs using a spectral response matrix $\mathbf{R}$. The original spatial resolution is preserved, while the spectral dimension is reduced from C-bands to three channels for CAVE and ICVL, and to four channels for PaviaU, Chikusei, and QUH-Qingyun. For the CAVE dataset, the original spatial resolution is preserved, containing 32 scenes, where 20 scenes are used for pretraining and the remaining 12 for testing. For the ICVL dataset, the data are randomly cropped to 512 × 512, where 30 out of 50 HSIs are used for training and 20 for testing. For the PaviaU, Chikusei, and QUH-Qingyun datasets, a 512 × 256 region is cropped from the top-left corner and evenly divided into training and testing subsets. To quantitatively assess the final reconstruction quality, we employ six widely used metrics: the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM), the spectral angle mapper (SAM), the Erreur Relative Globale Adimensionnelle de Synthése (ERGAS), the cross correlation (CC), and the no-reference natural image quality evaluator (NIQE). PSNR, SSIM, and CC evaluate spatial reconstruction quality, while SAM and ERGAS assess spectral fidelity and overall reconstruction accuracy. NIQE is used to evaluate perceptual quality without requiring ground-truth reference images.

2) Implementation Details: Our model is implemented in PyTorch and trained on a single NVIDIA GeForce RTX 3090 GPU in two stages. SDNet is implemented as a five-layer MLP with 512 hidden dimensions, SiLU activations, and a three-layer time-embedding module. It is pretrained on HSIs for 10^6 iterations with a batch size of 1024, a learning rate of 10^{-2}, and a diffusion process of 2000 steps. A separate SDNet is pretrained for each benchmark dataset using its training split, allowing it to capture the spectral distributions of different data sources. For the self-supervised optimization of WCAR-Net, we use the Adam optimizer and the Haar wavelet for multiscale decomposition [57]. The maximum number of iterations is $K = 4000$, with a learning rate of 0.005.

B. Comparison With State of the Art

1) Comparison Methods: To provide a comprehensive evaluation, we compare the proposed SDSP with several state-of-the-art methods across diverse paradigms.

1) *Model-Based Methods:* Arad [29].
2) *CNN-Based Methods:* HPRN [37], MSFN [38], FSDFF [35], and the unsupervised UnGUN [19].
3) *Transformer-Based Methods:* NukesFormers [21], which utilizes nonuniform domain alignment.
4) *Diffusion-Based Methods:* Lib-SRDM [22], a spectral library-aided method.
5) *Implicit Representation-Based Methods:* INR-HSISR [40].

In terms of training paradigms, HPRN, MSFN, FSDFF, and INR-HSISR are supervised models trained on paired datasets, whereas UnGUN, NukesFormers, and Lib-SRDM represent unpaired or unsupervised approaches. This broad selection of baselines ensures a rigorous evaluation against various technical architectures and supervision levels. The official codes were used for all baseline methods. Hyperparameters were set based on the original guidelines and fine-tuned to achieve the best possible performance.

2) Quantitative Comparison: The average PSNR, SSIM, SAM, ERGAS, CC, and NIQE results on the CAVE, ICVL, PaviaU, Chikusei, and QUH-Qingyun datasets are summarized in Table I. We first evaluate SDSP using average quantitative metrics. Across the five benchmark datasets, it consistently delivers competitive or superior performance. On CAVE, it achieves the highest PSNR (36.45), SSIM (0.977), and CC (0.993), as well as the lowest ERGAS (4.46), demonstrating strong spatial and spectral fidelity. Similar trends are observed on ICVL, Chikusei, and QUH-Qingyun, where SDSP achieves the best performance on most metrics while remaining among the top-performing methods on PaviaU. Notably, SDSP performs comparably to or better than supervised methods such as HPRN and MSFN, despite using no paired RGB–HSI supervision during reconstruction. These results validate the proposed SDSP, whose learned diffusion guidance enables high-quality self-supervised HSI reconstruction.

3) Visual Comparison: We present visual comparisons on the CAVE, ICVL, PaviaU, Chikusei, and QUH-Qingyun datasets in Fig. 3. The odd rows show the reconstructed false-color images obtained by different methods, while the even rows present the corresponding residual maps, computed as the average absolute difference across selected spectral bands. These results indicate that our method achieves more accurate reconstruction with lower residual errors. Specifically, closer examination of the spatial reconstruction results in Fig. 3 shows that other methods generally suffer from artifacts. Some convolution-based methods, such as FSDFF, introduce checkerboard artifacts that disrupt the natural appearance of the image, while UnGUN and INR-HSISR exhibit severe discontinuities at object boundaries. In contrast, our method preserves image continuity, resulting in smoother transitions

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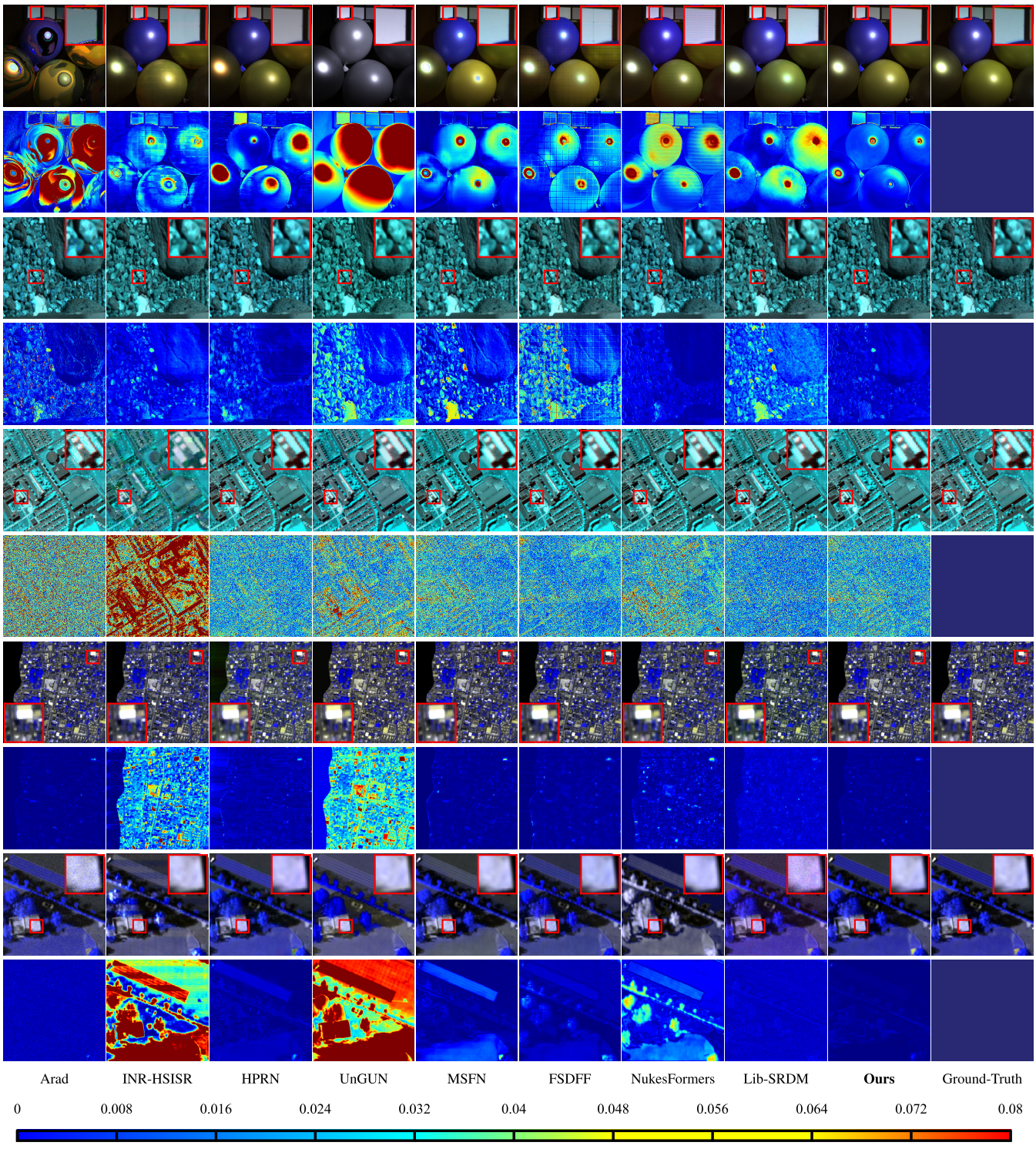


Fig. 3. Visualization results and corresponding error maps for the Balloons scene from the CAVE dataset, the BGU scene from the ICVL dataset, the PaviaU scene, the Chikusei scene, and the QUH-Qingyun scene, respectively. For each scene, the odd-numbered rows present the reconstructed images, while the even-numbered rows show the corresponding error maps.

between objects and the background while faithfully restoring fine textures, such as reflections in the Balloons scene. The error maps further confirm this result, with our method consistently showing minimal deviation (uniformly blue regions), indicating more faithful overall spatial reconstruction.

Furthermore, the importance of spectral fidelity should not be overlooked. To illustrate this, we include spectral profile comparisons in Fig. 4. The spectral profiles, which represent the pixel value distribution across different wavelengths, provide valuable insights into how well each method

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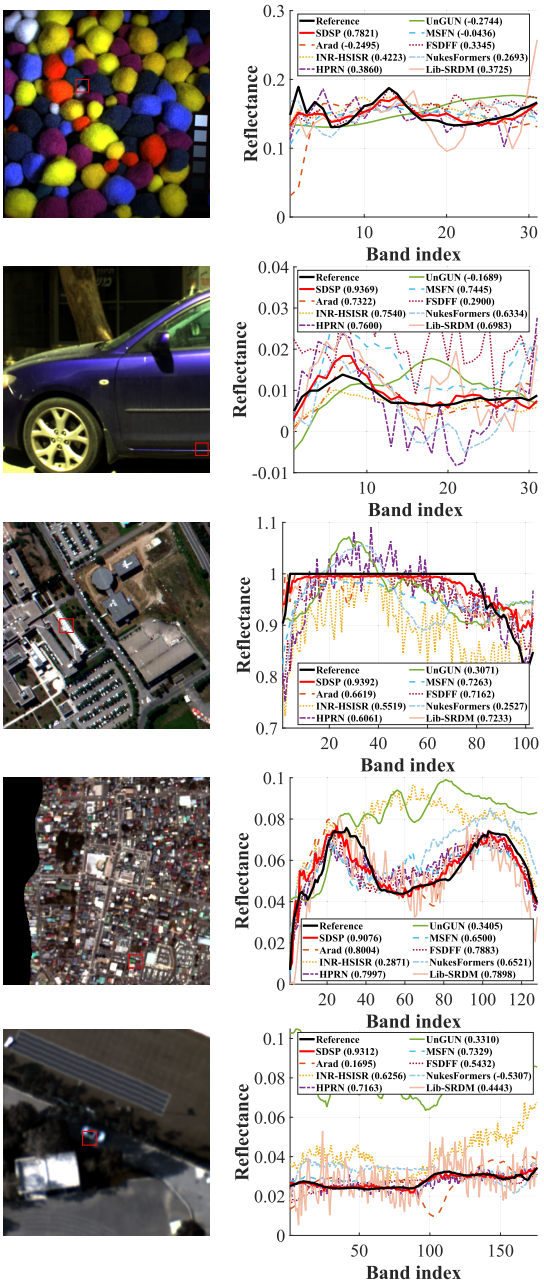


Fig. 4. Spectral curve comparison for a selected region on each dataset. (Top to bottom) CAVE ("Pompoms"), ICVL ("Sat_0406–1107"), PaviaU, Chikusei, and QUH-Qingyun. The selected region is indicated by a red box, and (right) average spectral curves within that region reconstructed by different methods. Correlation coefficients (corr.) are included in the legend.

preserves spectral information. Upon examination, it is evident that the spectral profiles generated by our proposed SDSP framework closely align with the ground truth, with its superior correlation coefficients providing quantitative support. This alignment strongly indicates our technique's excellent spectral preservation ability. In contrast, competing methods often fail to capture the spectral shape. A notable example is the negatively correlated spectrum from UnGUN on the CAVE dataset, which is physically implausible. In other words, our method excels in reproducing the original spectral characteristics, highlighting the crucial role of our score-distilled spectral expert in ensuring accurate spectral reconstruction.

C. Ablation Study

1) Ablation Study on SDNet and WCAR-Net: The proposed method comprises two key components: SDNet, providing a data-driven spectral prior, and WCAR-Net, serving as a spatial–wavelet deep prior. Together, they jointly regularize reconstruction from complementary spectral and spatial perspectives. To assess their necessity and complementarity, we perform ablation studies by removing each module individually, denoted as w/o SDNet and w/o WCAR-Net. As reported in Table II, removing SDNet and relying solely on WCAR-Net causes a severe PSNR drop from 36.45 to 20.16 dB, showing that spatial modeling alone cannot compensate for the lack of spectral guidance. Conversely, replacing WCAR-Net with a basic learnable tensor reduces the PSNR to 26.19 dB, suggesting that spectral prior alone cannot ensure accurate spatial structure reconstruction. These results demonstrate that SDNet and WCAR-Net are not only complementary but also mutually reinforcing, where SDNet provides spectral guidance to improve spatial reconstruction, while WCAR-Net enhances spatial detail recovery, and their joint optimization yields progressive improvements in spatial detail and spectral fidelity.

2) Ablation Study on CAR and Wavelet Transform: To evaluate the contributions of key modules within WCAR-Net, namely the CAR module and wavelet transform, we conduct additional ablation studies, with results reported in Table III. Replacing the wavelet transform with pooling and bicubic interpolation, and substituting the CAR module with common concatenation or summation, all led to noticeable performance drops. In addition, we perform ablation studies on the three internal components of the CAR module, namely GatedConv, SE, and SA, to investigate their respective effects on performance. These results confirm that the wavelet transform and CAR module are essential for achieving superior performance.

3) Ablation Study on the Pretraining Data Quantity: We analyze the impact of SDNet pretraining from two aspects: dataset size and spatial coverage. First, we evaluate the effect of dataset size on the CAVE dataset by training SDNet with 1–20 HSI scenes, as summarized in Table IV. As the number of training scenes increases, the reconstruction performance gradually improves and becomes more stable. The relatively small performance gap between using 1 and 20 scenes can be attributed to the high data efficiency of the proposed 1-D spectral learning strategy, since each CAVE image already provides 262 144 (512×512) pixelwise spectral samples for learning the spectral prior. Furthermore, we investigate the influence of training spatial coverage during pretraining on the QUH-Qingyun dataset. The results are reported in Table V, where the reported ratios denote the fraction of the spatial side length used for training. The 1.00 setting provides 65 536 (256×256) pixelwise spectral samples, whereas the 0.25 setting provides only 4096 (64×64) samples. Consequently, reducing the pretraining area leads to a much more pronounced performance degradation, indicating that sufficient pixelwise spectral samples are essential for learning a high-quality spectral prior. Based on these observations,

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TABLE III
ABLATION STUDY RESULTS OF CAR AND WAVELET TRANSFORM ON CAVE AND QUH-QINGYUN DATASETS

Dataset	Metric	CAR→ ⊕	CAR→ Concat	w/o Wavelet	w/o GatedConv	w/o SE	w/o SA	WCAR-Net
CAVE	PSNR ↑	35.23	35.51	36.00	36.00	35.53	36.13	**36.45**
	SSIM ↑	0.971	0.969	0.976	**0.977**	0.976	**0.977**	**0.977**
	SAM ↓	13.23	13.56	11.53	11.40	11.40	11.23	**11.15**
	ERGAS ↓	5.10	4.93	4.78	4.98	5.38	4.76	**4.46**
	CC ↑	0.992	0.992	0.992	0.992	0.991	0.992	**0.993**
	NIQE ↓	0.159	0.152	0.144	0.144	0.150	**0.143**	**0.143**
QUH-Qingyun	PSNR ↑	43.57	43.34	43.50	41.76	37.47	41.54	**44.58**
	SSIM ↑	0.990	0.991	0.990	0.988	0.988	0.987	**0.991**
	SAM ↓	2.46	2.46	2.31	2.61	2.62	2.88	**2.01**
	ERGAS ↓	0.85	0.81	0.81	2.09	3.77	2.21	**0.77**
	CC ↑	**0.995**	**0.995**	**0.995**	0.993	0.977	0.992	**0.995**
	NIQE ↓	**0.154**	0.156	**0.154**	0.157	0.162	0.163	0.159

TABLE IV
ABLATION STUDY RESULTS OF PRETRAINING DATASET SIZE ON CAVE

Pre-training data size	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	NIQE ↓
1	34.94	0.973	12.66	6.18	0.990	0.135
5	35.86	0.975	11.92	5.32	0.990	0.140
10	35.93	0.976	11.45	5.96	0.991	**0.134**
15	36.27	**0.977**	11.31	4.75	0.992	0.143
20	**36.45**	**0.977**	**11.15**	**4.46**	**0.993**	0.143

TABLE V
ABLATION STUDY RESULTS OF PRETRAINING AREA FRACTION ON QUH-QINGYUN

Area fraction	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	NIQE ↓
0.25	38.50	0.979	4.49	1.52	0.984	0.157
0.50	39.33	0.982	4.61	1.41	0.986	**0.147**
0.75	41.34	0.989	3.28	1.08	0.992	0.156
1.00	**44.58**	**0.991**	**2.01**	**0.77**	**0.995**	0.159

TABLE VI
ABLATION STUDY RESULTS OF DIFFERENT WAVELET BASES ON QUH-QINGYUN AND CAVE DATASETS

Dataset	Wavelet	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓	CC ↑	NIQE ↓
CAVE	db2	36.20	0.973	13.53	2.32	**0.993**	**0.140**
	sym4	**36.50**	**0.977**	11.58	**2.26**	**0.993**	0.142
	bior4.4	36.23	0.974	12.27	2.37	0.992	0.145
	Haar	36.45	**0.977**	**11.15**	4.46	**0.993**	0.143
QUH-Qingyun	db2	44.23	0.990	2.08	0.79	0.994	0.162
	sym4	44.20	0.990	2.09	0.80	0.994	**0.159**
	bior4.4	44.34	0.990	2.04	0.78	**0.995**	0.160
	Haar	**44.57**	**0.991**	**2.01**	**0.77**	**0.995**	**0.159**

SDNet pretrained on the entire training spatial domain or all 20 scenes is adopted in all subsequent experiments to ensure stable reconstruction performance.

4) Ablation Study on Different Wavelet Bases: To further evaluate the robustness of WCAR-Net, we compare the Haar wavelet with several representative bases: Daubechies-2 (db2) [58], Symlets-4 (sym4) [58], and Biorthogonal-4.4 (bior4.4) [59]. As reported in Table VI, the proposed framework achieves consistent and high-quality reconstruction performance across all tested wavelet filters. The Haar wavelet is retained as the default choice because it provides an optimal tradeoff between reconstruction accuracy and computational complexity during the self-supervised optimization process.

D. Computational Efficiency and Complexity

To evaluate the practical applicability and computational efficiency of the proposed method, we compare GFLOPs, parameter size, training/pretraining time, and inference time of all competing methods on the QUH-Qingyun dataset, as summarized in Table VII. Although the proposed self-supervised framework introduces additional test-time optimization compared with feedforward supervised methods, it benefits from an efficient 1-D pretraining strategy, resulting in substantially lower pretraining cost than most competing methods. The longer inference time is mainly due to the image-specific optimization of WCAR-Net for each test image under the guidance of the pretrained SDNet. Unlike existing methods that directly apply fixed network parameters during inference, the proposed framework adapts WCAR-Net to each input image, providing greater flexibility for image-specific reconstruction. We further observe that HPRN achieves the shortest training time and very fast inference while maintaining competitive reconstruction performance. Similarly, MSFN achieves performance comparable to the proposed method with relatively low computational cost. However, both methods are supervised and rely on paired RGB–HSI training data, which are typically difficult and expensive to acquire in real-world applications. Among the unsupervised methods, UnGUN achieves the lowest overall computational cost but suffers from inferior reconstruction performance, whereas LiteSRDM and NukesFormers require substantially higher training or pretraining costs. In contrast, the proposed method achieves the best reconstruction performance among the self-supervised and unsupervised methods while maintaining a relatively moderate pretraining complexity.

E. Discussion on Advantages and Limitations

The proposed self-supervised SDSP framework eliminates the need for paired RGB–HSI datasets. By learning a continuous generative model of spectral distributions from unpaired

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TABLE VII
COMPARISON OF COMPUTATIONAL EFFICIENCY AND COMPLEXITY ON THE QUH-QINGYUN DATASET

Category	Method	GFLOPs	Params (M)	Train / Pre-train Time	Test Time	PSNR	SSIM	SAM	ERGAS	CC	NIQE
Self-supervised	Ours	667.2	42.1	1.48 h	476 s	**44.58**	**0.991**	**2.01**	**0.77**	**0.995**	0.159
Unsupervised	Lib-SRDM	2445.2	95.2	7.19 h	82 s	41.51	0.963	8.68	4.68	0.943	0.296
	NukesFormers	820.8	33.5	19.66 h	8 s	33.14	0.914	14.60	11.35	0.672	0.146
	UnGUN	74.4	0.6	0.55 h	2 s	27.97	0.928	9.50	7.33	0.901	0.154
Supervised	FSDFF	22562.4	552.8	12.33 h	9 s	34.74	0.963	4.72	3.69	0.948	0.217
	MSFN	2018.7	77.9	1.51 h	10 s	44.11	**0.991**	4.41	3.38	0.964	0.100
	HPRN	2035.2	15.6	0.11 h	4 s	43.53	0.987	5.37	3.54	0.962	**0.082**
	INR-HSISR	605.3	73.5	7.2 h	1 s	29.04	0.894	9.73	9.86	0.893	0.212

hyperspectral data, SDSP introduces a diffusion-based spectral prior into the reconstruction process via SDS, guiding the self-supervised optimization of WCAR-Net. Unlike existing methods that rely on explicit data anchoring, such as reference images or spectral libraries, the proposed framework provides a learned spectral prior that enables spectrally faithful reconstruction while improving robustness to spectral variations. In addition, the pixelwise formulation of SDNet enables efficient prior learning from a limited number of clean hyperspectral samples. However, due to the image-specific self-supervised optimization adopted during inference, the proposed method requires longer inference time than feedforward models, which may limit its applicability in real-time scenarios. Furthermore, the learned spectral prior may be affected by domain variations; when the spectral distribution deviates significantly from the pretraining data, reconstruction performance may degrade. In future work, we will focus on improving inference efficiency and enhancing the generalization capability of the learned spectral prior.

V. CONCLUSION

In this work, we propose SDSP, a self-supervised framework for hyperspectral super-resolution from a single RGB observation. By integrating a WCAR-Net with a pretrained spectral diffusion model, our method effectively combines spatial and spectral deep priors to tackle the highly ill-posed RGB-to-HSI problem. The diffusion model provides spectral guidance via SDS, while WCAR-Net fuses spatial details with high-level semantic features and preserves fine frequency-domain structures through wavelet-based decomposition. Trained iteratively in a self-supervised manner, SDSP eliminates the need for costly paired RGB–HSI data. Extensive experiments on multiple benchmark datasets demonstrate that our method achieves high-fidelity reconstructions, outperforming state-of-the-art approaches in both spectral accuracy and spatial quality, highlighting its robustness and practical applicability in real-world scenarios with limited HSI.

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